Exp: 8.7 Flow Control at Data link layer using Sliding Window Protocol

Aim: To write a program to implement flow control at the Data link layer using the Sliding Window Protocol, and to simulate the flow of frames from one node to another.

Algorithm:

1. Start the program.
2. Read the total number of frames to be sent (N), the sender's window size (W) and a frame number to simulate as lost during transmission
3. Repeat while base < N:
   a. Determine the current sender window as the range [base, base+W-1].
   b. Send all frames with sequence numbers from nextSeq up to , incrementing nextSeq after each frame, while the window restricts the sender to at most W outstanding
4. If the lost frame lies within the frames just sent:
   i) The receiver discards the lost frame and any subsequent out of order frames.
5. Repeat step 4 until all N frames have been transmitted and acknowledged
6. Stop the program.

OUTPUT

Enter total number of frames to send: 8
Enter sender's window size: 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss) : 5

Sender Window : [Frame 0 - Frame 3]

Node A -> Sending Frame 0
Node B -> Received Sended frame 4
Node A -> Timeout! resending window starting from Frame 5

Sender Window : [Frame 4 - Frame 7]

Node A -> Sending Frame 5
Node A -> Sending Frame 6
Node B -> Received Frame 5, sending ACK 5
Node A -> ACK 7 received sliding window Forward

All 8 frames Successfully written in Node A to Node B

Program:

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0

    while base < total_frames:
        window_end = min(base + window_size - 1, total_frames - 1)
        print(f"Sender Window : [Frame {base} - Frame {window_end}]")
    while next_seq < base + window_size and next_seq < total_frames:
        print(f"Node A -> Sending Frame {next_seq}")
        if next_seq == lost_frame:
            print(f"  (Frame {next_seq} lost in transit!)")
        next_seq += 1

    if lost_frame is not None and base <= lost_frame < next_seq:
        print(f"Node B -> Frame {lost_frame} missing. Discarding out-of-order
frames after it.")
        print(f"Node A -> Time out! Resending window starting from
                                                    Frame {lost_frame}
                                                    ")
    print("\n---")
print(f"All {total_frames} frames successfully transmitted from Node
A to Node B.")
def main():
    print("Sliding window protocol - flow control simulation")
    total_frames = int(input("Enter total number of frames to send :"))
    window_size = int(input("Enter sender window size :"))
    ~~lost_frame~~
    if lost_frame == -1:
        lost_frame = None
```

Result:

The program to implement flow control at the Data link layer using the Sliding window Protocol was successfully written in Python, executed and verified by simulating the flow of frames between two nodes